

GUREHMAT CHAHAL

Toronto, ON

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Education

McMaster University

Honours Bachelor of Applied Science, Computer Science (Co-op)

2025 – 2029

Hamilton, ON

Experience

AI Programmer

June 2026 – Present

Prepr Foundation

Toronto, ON (Remote)

- Eliminated a **P1 production security vulnerability** exposing user PII to unauthenticated requests, closing the leak across **11 Laravel API resource files** by gating serialized fields behind `auth('api')->check()` checks.
- Reduced duplicated validation logic across a **1,200+ file PHP/Laravel** codebase into a single source of truth by extracting reusable **Rule classes** for name, email, and soft-delete-aware existence checks.
- Unblocked local development for a **4-person team** by making database seeders idempotent to fix recurring Docker crash-loops, and surfaced OAuth open-redirect and stored XSS findings in a security audit.

AI / Backend Developer

May – July 2026

Aeternum

Remote

- Guaranteed zero unauthorized data exposure in a **RAG search pipeline** over private user records by enforcing per-record access control *before* retrieval, using a GIN-indexed array-overlap pre-filter so restricted rows never reach vector ranking.
- Cut near-miss noise from search results by tuning an empirical **0.78 similarity floor** over **e5-base-v2** embeddings ranked through **pgvector HNSW**.
- Delivered **all 14 remaining backend tickets** as sole developer after a co-intern's mid-project departure, shipping **FastAPI/PostgreSQL** search and RAG-answer endpoints through branch-per-ticket PR review.

Workflow & Integration Lead

Apr – June 2026

WareShare

Remote

- Prevented duplicate charges and duplicate customer notifications in production by designing and end-to-end validating an **idempotent booking-to-payment state machine** spanning agreement negotiation, **Stripe** checkout, and webhook confirmation on **Hono/Cloudflare Workers**.
- Raised transactional email coverage from **7 to 16 of 16** specified flows by architecting a **19-trigger** notification layer (**Resend**) across signup, booking, and invoicing, feature-flagged for safe rollout.
- Enabled accurate real-time availability across concurrent bookings by building a capacity-aware reservation engine that reserves space across three booking statuses and drives calendar display.

AI Developer (Team Lead)

Feb – Mar 2026

Workplace Edge

Remote

- Shipped a production AI coaching chatbot for a live eLearning client in **4 weeks** by leading a **3-person team**, owning the **Express/TypeScript API**, **Turso (libSQL)** data layer via Prisma, and **Gemini 2.5 Flash** integration.
- Kept responses grounded in client curriculum rather than generic AI advice by injecting all **20 lesson plans** into Gemini's **1M-token context**, removing the need for a vector database.
- Constrained off-scope and unsafe output with a **4-category intent router** and a PII-stripping safety layer applied before persistence.

Founder & CTO

2024 – 2025

Advancing Future Innovators (AFI)

Ontario

- Delivered programming education to **30+ students per session** by founding a nonprofit and building school partnerships, running operations and volunteer coordination across multiple local schools.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, PHP, C, SQL, HTML/CSS, Haskell

Frameworks: React 18/19, FastAPI, Hono, Express, Node.js, Laravel, Prisma, Tailwind CSS, Vite

Data & Infra: PostgreSQL, pgvector, Turso (libSQL), MongoDB Atlas, Supabase, Cloudflare Workers/D1/R2, Vercel, Docker

Tools & AI: Git, GitHub Actions, Stripe, Jira, Cursor, Gemini API, RAG, Vector Embeddings, Prompt Engineering

Leadership / Extracurricular

NASA Space Settlement Design Contest – Grand Prize (9,000+ participants, 12 countries). **Hackathons:** The Record (Best Use of ElevenLabs), SEED (Best UI), C.A.P.T.C.H.A. (solo). **Also:** MEC Debate (2nd), Mac AI Club, EWB, Socratica.